

Answer Key — Biology Grade 11

Sub-Strand 1.4: Cell Division • Lesson 2: What Is Cell Division? Meaning of Cell Division and the Structure of the Chromosome

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	D	B	A	D	C	C

USE AFTER: OBSERVE PHASE — Activity 1: observation table of cells before and after division

Q1. A group compares printed pictures of a cell before and after division. Before division they count 1 cell, and after division they count 2 cells. Which conclusion is best supported by this evidence?

- A. Food that we eat turns directly into new cells
- B. One existing cell divides to make new cells**
- C. A cell only gets bigger and bigger without making new cells
- D. New cells are brought into the body from outside, one by one

Correct answer: B. The counts show 1 cell becoming 2, so the evidence supports the idea that an existing cell divides to form new cells. The other ideas are common predictions that the counts do not support.

USE AFTER: OBSERVE PHASE — Activity 1: observation table (1 → 2, and 2 → 4 if shown)

Q2. In a growing child, one cell divides into 2 cells. Then each of those 2 cells divides once. How many cells are there after the second round of division?

- A. 6
- B. 8
- C. 3
- D. 4**

Correct answer: D. Each division doubles the number of cells: $1 \rightarrow 2 \rightarrow 2 \times 2 = 4$ cells. Choosing 3 comes from adding one cell instead of doubling.

USE AFTER: OBSERVE PHASE — Activity 2: clay model of 1 cell → division → 2 cells

Q3. A group builds a clay model of one cell dividing into two cells. They put a nucleus in the original cell. What must the two new cells receive so that they can work like the original cell?

- A. Only extra food, because the new cells have no need for anything else
- B. A copy of the genetic information (instructions) found in the nucleus**
- C. Nothing, because each new cell makes its own instructions from scratch
- D. Only a bigger cell size than the original cell

Correct answer: B. New cells can only function correctly if they receive the genetic information (instructions) from the parent cell's nucleus. Food and size alone do not tell a cell how to work.

USE AFTER: EXPLAIN PHASE — Activity 3: Jigsaw and closing cold-calls on the meaning of cell division

Q4. Which statement best gives the meaning of cell division?

- A. A parent cell dividing to form new daughter cells**
- B. A cell becoming bigger without changing in number
- C. Two chromosomes joining to form one cell
- D. Food being changed into new cells

Correct answer: A. Cell division is the process by which one parent cell divides to form new daughter cells. Growing bigger or food turning into cells does not describe division.

USE AFTER: EXPLAIN PHASE — Activity 3: Jigsaw, Expert B (the centromere) and labelling the chromosome diagram

Q5. A chromosome has two identical arms that are joined together at one point. What is the name and job of the part where the arms are joined?

- A. A small separate cell that makes the chromosome
- B. The cell membrane, which keeps the chromosome inside the cell
- C. The gene, which is a second chromosome inside the first
- D. The centromere, which holds the two chromatids together**

Correct answer: D. The centromere is the region where the two chromatids are joined. It is a part of the chromosome, not a cell or a separate chromosome.

USE AFTER: EXPLAIN PHASE — Activity 3: Jigsaw, Experts A and F (the chromatid, arms and sister chromatids)

Q6. A chromosome is made of two identical arms joined at the centromere. What are these two identical arms called?

- A. Two separate cells
- B. Two different chromosomes
- C. Two sister chromatids**
- D. Two centromeres

Correct answer: C. The two identical arms of one chromosome are sister chromatids. They are not two different chromosomes.

USE AFTER: END OF LESSON — End of lesson: exit ticket on labelling a chromosome and linking chromosomes to genetic information

Q7. Humans have 46 chromosomes in their cells and maize has 20. Which statement about chromosomes and DNA is correct?

- A. Chromosomes are made of cells, and DNA is a type of chromosome
- B. Chromosomes and DNA are two completely unrelated parts of a cell
- C. Chromosomes contain DNA, and genes on the DNA carry the genetic information a new cell needs**
- D. Only humans have chromosomes, and maize has only DNA

Correct answer: C. A chromosome contains DNA, and genes are regions of DNA that carry instructions. The different numbers (46 in humans, 20 in maize) show that different organisms have different chromosome numbers, but both have chromosomes.